

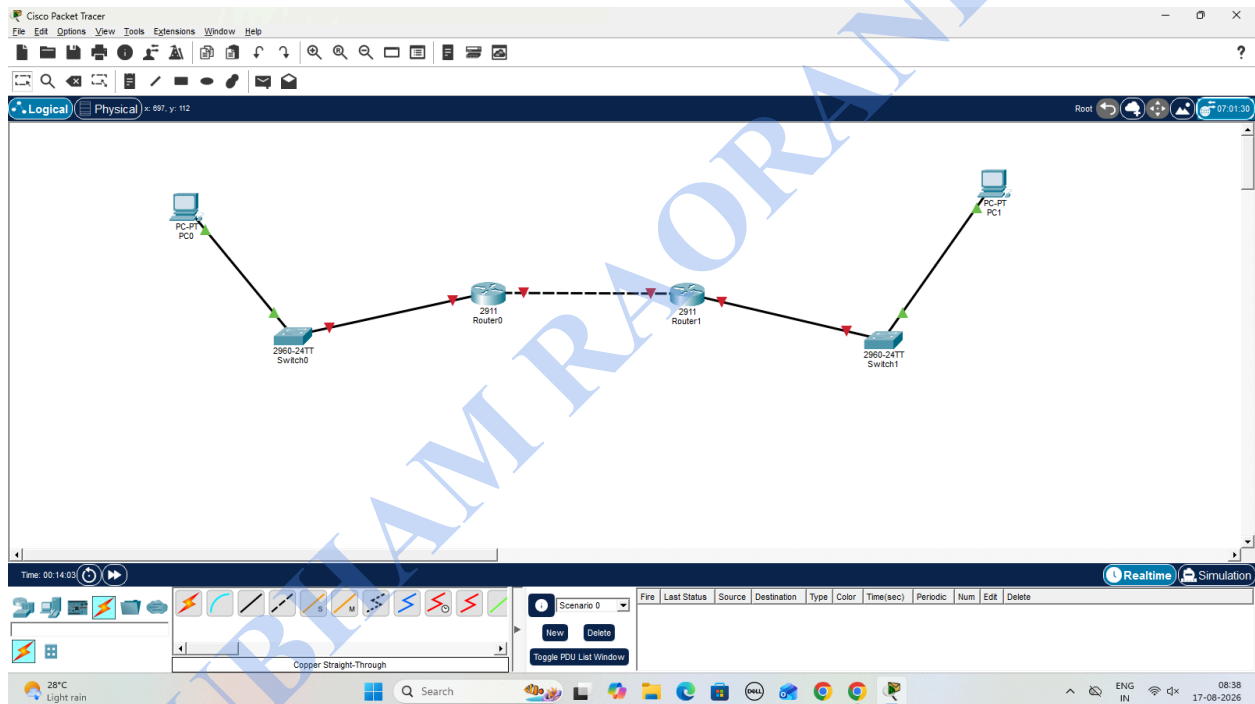
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SUBJECT : CYBER SECURITY

AIM : Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Devices Required

- 2 Cisco 2911 routers
- 2 Cisco 2960 switches
- 2 PCs
- Copper Straight-Through cables
- Copper Cross-Over cable



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Procedure

Step 1: Open Cisco Packet Tracer

1. Open Cisco Packet Tracer.
2. Place two Cisco 2911 routers.
3. Place two Cisco 2960 switches.
4. Place two PCs.
5. Rename the routers as R1 and R2.
6. Rename the switches as SW1 and SW2.
7. Rename the PCs as PC1 and PC2.

Step 2: Connect the Devices

Connect the devices as follows:

PC1 FastEthernet0 → SW1 FastEthernet0/1
SW1 GigabitEthernet0/1 → R1 GigabitEthernet0/1
R1 GigabitEthernet0/0 → R2 GigabitEthernet0/0
R2 GigabitEthernet0/1 → SW2 GigabitEthernet0/1
SW2 FastEthernet0/1 → PC2 FastEthernet0

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Configure PC0

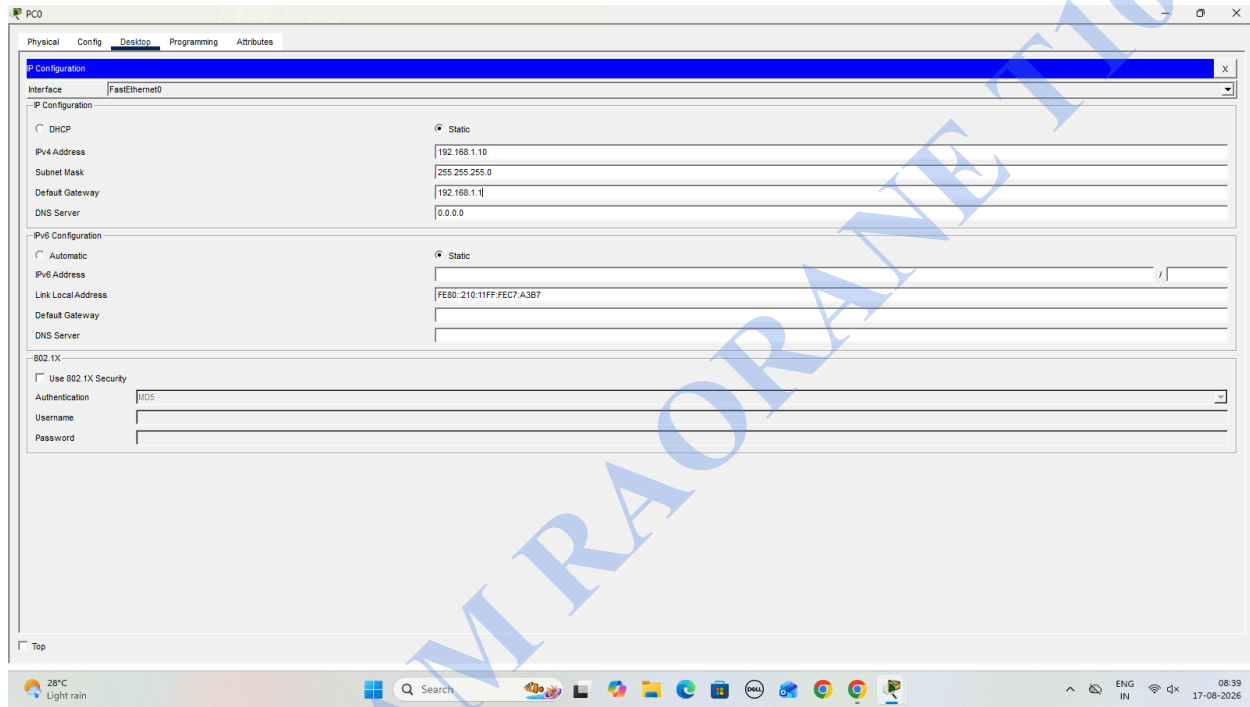
Click PC1 → Desktop → IP Configuration.

Enter:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1



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Configure PC1

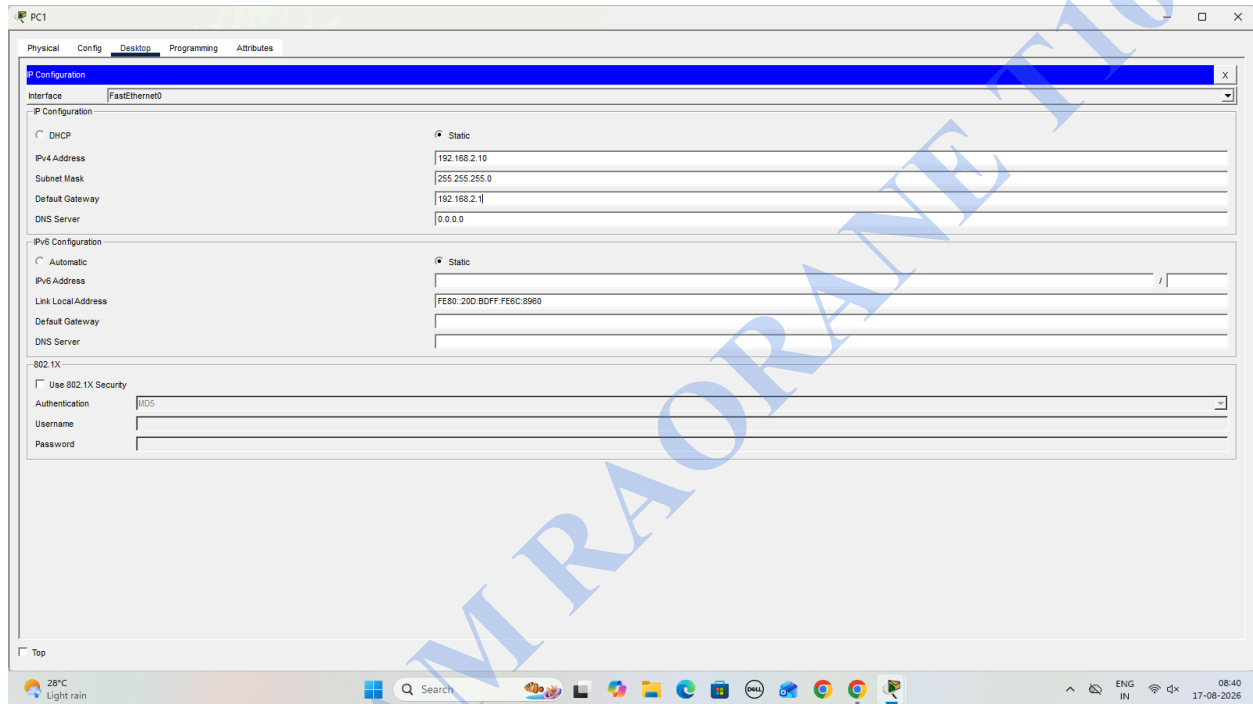
Click PC2 → Desktop → IP Configuration.

Enter:

IP Address: 192.168.2.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1



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ROUTER 0 CONFIG

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

R1#end
Translating "end"...domain server (255.255.255.255)
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#show interfaces serial
% Incomplete command.
R1#
% Unknown command or computer name, or unable to find computer address

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface serial 0/0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#no shutdown

%LINK-3-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#clock rate 64000
R1(config-if)#interface serial 0/0/1
R1(config-if)#ip address 10.0.0.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
%LINK-3-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
R1(config)#show ip interface brief
% Invalid input detected at '' marker.

R1(config)#enable
% Incomplete command.
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 10.0.0.1 YES manual up
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#
```

ROUTER1 CONFIG

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 unassigned YES unset administratively down down
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
Router(config)#hostname R2
R2(config)#interface gigabitEthernet 0/0
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#interface serial 0/0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252ip address 10.0.0.2 255.255.255.252
% Invalid input detected at '' marker.
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shutdown

%LINK-3-CHANGED: Interface Serial0/0/0, changed state to down
R2(config-if)#exit
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

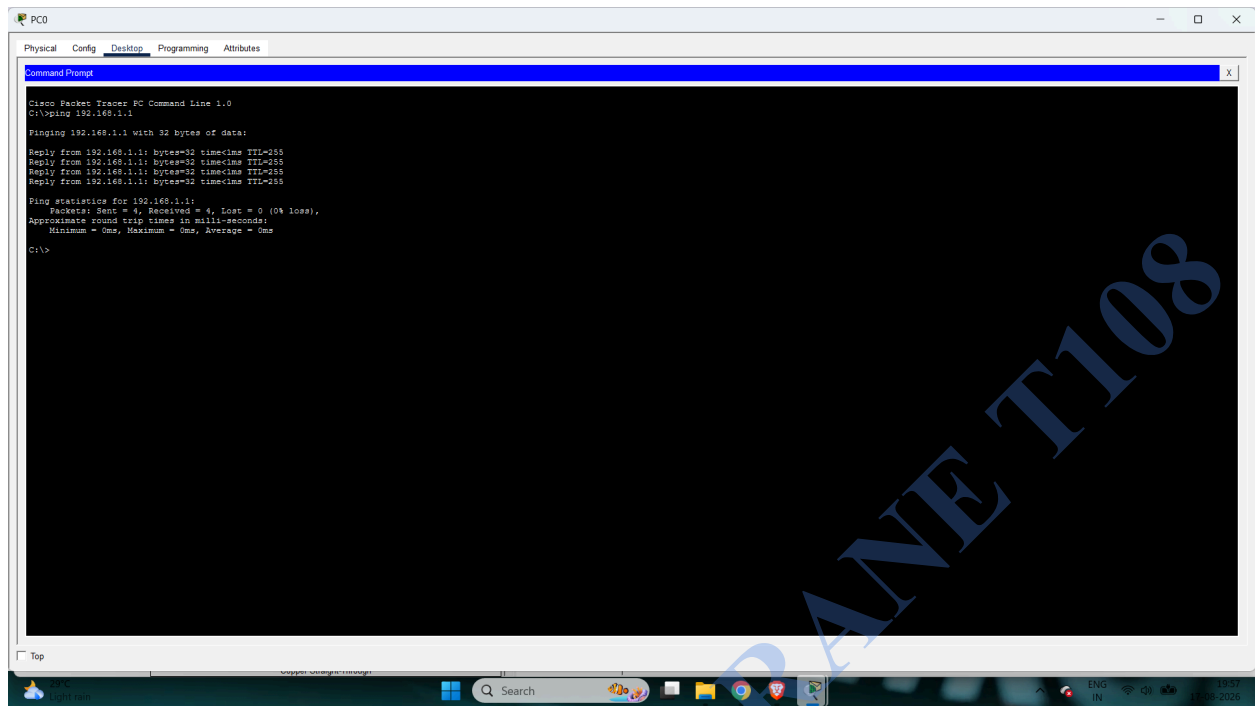
R2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.2.1 YES manual up down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 10.0.0.2 YES manual down
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R2#
%LINK-3-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: line protocol on Interface Serial0/0/0, changed state to up

R2#enable
R2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.2.1 YES manual up down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Serial0/0/0 10.0.0.2 YES manual up
Serial0/0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R2#
```

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BASIC NETWORK CONNECTIVITY - PC0



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC0. The command prompt displays the output of a ping command to 192.168.1.1. The output shows four successful replies with 32 bytes of data and a TTL of 255. The ping statistics indicate that 4 packets were sent, 4 were received, and there was 0% loss. The approximate round trip times in milliseconds are: Minimum = 0ms, Maximum = 0ms, and Average = 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

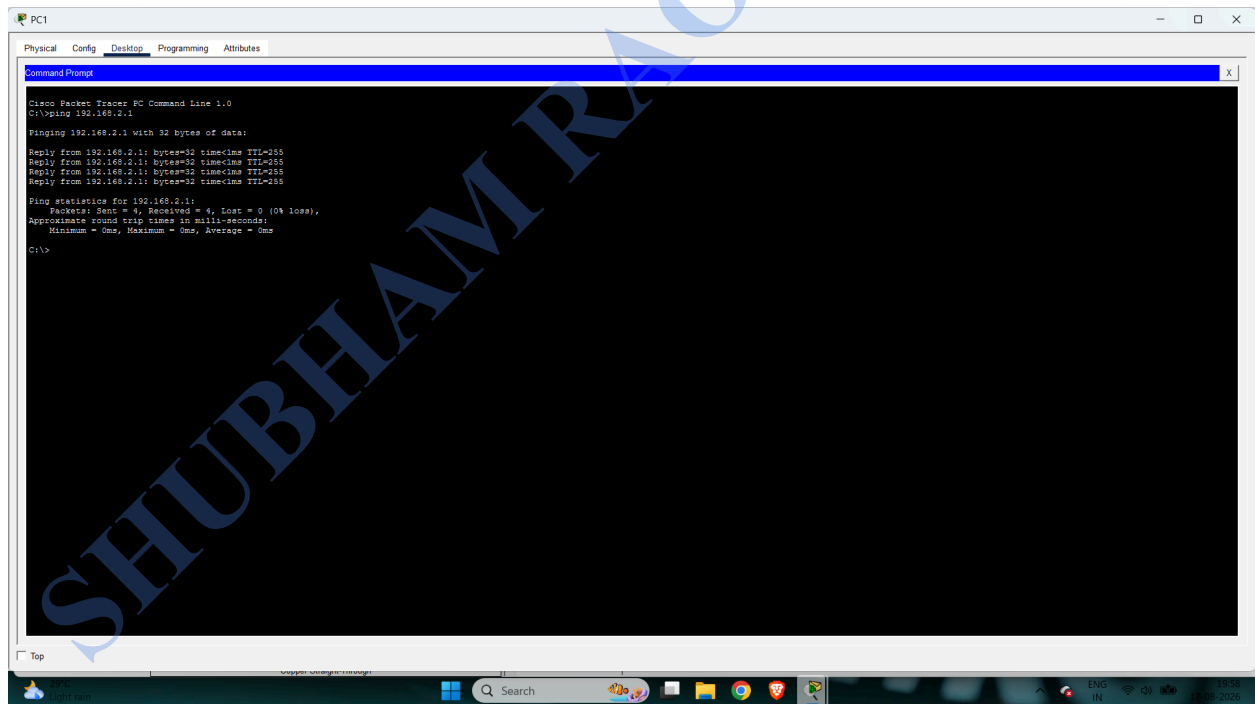
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC1



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC1. The command prompt displays the output of a ping command to 192.168.2.1. The output shows four successful replies with 32 bytes of data and a TTL of 255. The ping statistics indicate that 4 packets were sent, 4 were received, and there was 0% loss. The approximate round trip times in milliseconds are: Minimum = 0ms, Maximum = 0ms, and Average = 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=0ms TTL=255
Reply from 192.168.2.1: bytes=32 time=0ms TTL=255
Reply from 192.168.2.1: bytes=32 time=0ms TTL=255
Reply from 192.168.2.1: bytes=32 time=0ms TTL=255

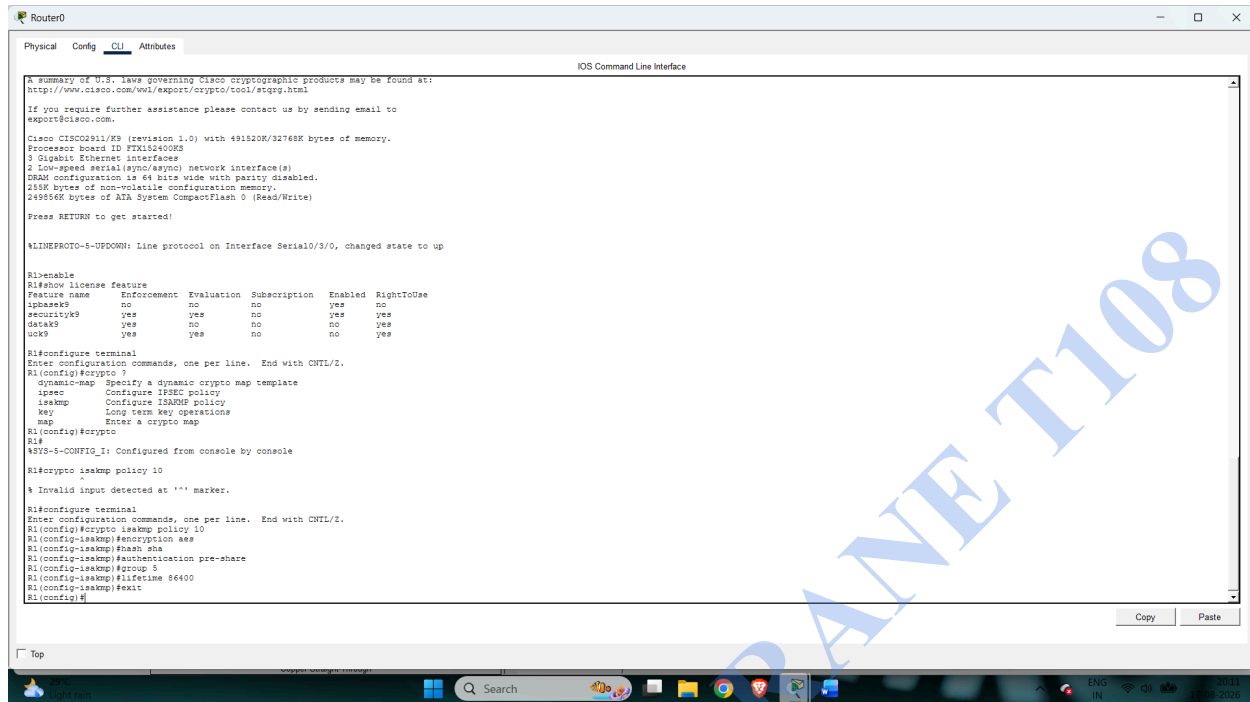
Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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ISKAMP ROUTER 0



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wll/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO02911/99 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400K3
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249956K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

$LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

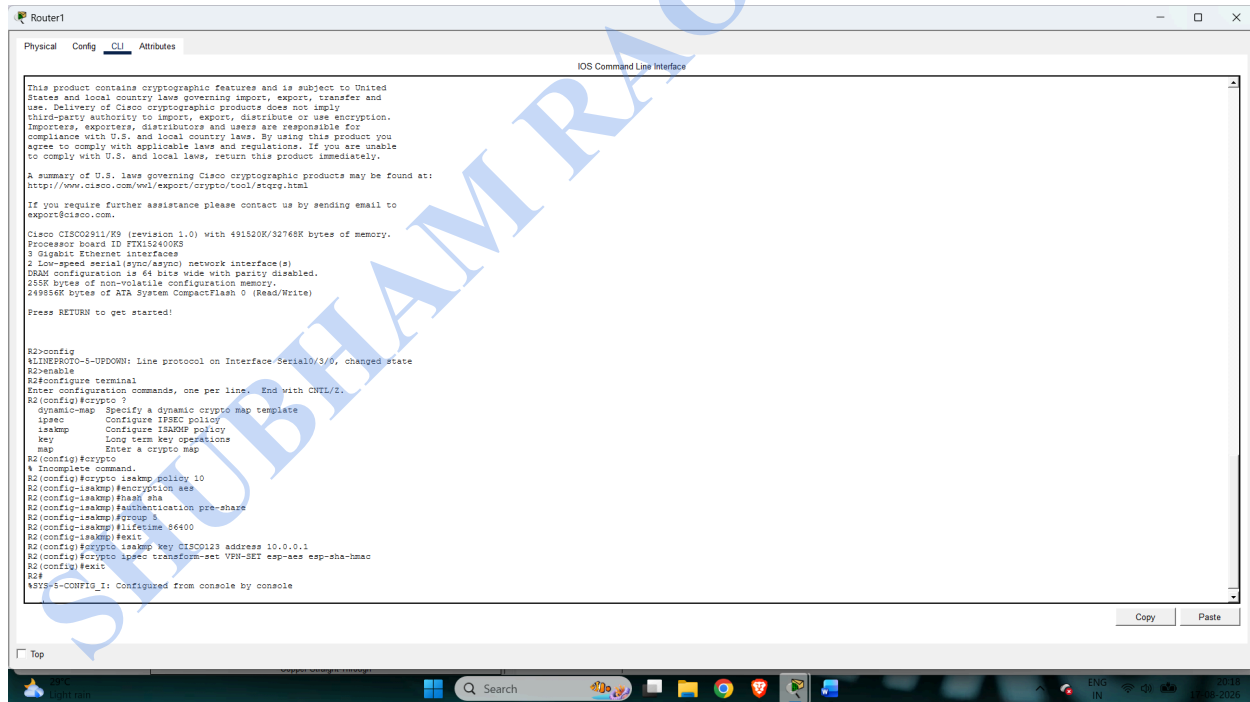
R1>enable
R1#show license feature
Feature name      Enforcement  Evaluation  Subscription  Enabled  RightToUse
-----
ipbase5          no           no          no             yes      no
securityk9       yes          yes         no             yes      yes
datak9           yes          no          no             no       yes
uclk9            yes          yes         no             no       yes

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto ?
dynamic-map      Specify a dynamic crypto map template
ipsec            Configure IPSEC policy
isakmp           Configure ISAKMP policy
key              Long term key operations
map              Enter a crypto map
R1(config)#crypto
R1#
$SYS-5-CONFIG_I: Configured from console by console

R1#crypto isakmp policy 10
% Invalid input detected at '' marker.

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#auth sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#lifetime 86400
R1(config-isakmp)#exit
R1(config)#
```

ROUTER 1



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wll/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO02911/99 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTK152400K3
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249956K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

R2>config
$LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#crypto ?
dynamic-map      Specify a dynamic crypto map template
ipsec            Configure IPSEC policy
isakmp           Configure ISAKMP policy
key              Long term key operations
map              Enter a crypto map
R2(config)#crypto
R2#
% Incomplete command.
R2(config)#crypto isakmp policy 10
R2(config-isakmp)#encryption aes
R2(config-isakmp)#auth sha
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 5
R2(config-isakmp)#lifetime 86400
R2(config-isakmp)#exit
R2(config)#crypto isakmp key CISCO123 address 10.0.0.1
R2(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R2(config)#exit
R2#
$SYS-5-CONFIG_I: Configured from console by console
```

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IPSEC TRANSFORM SET ROUTER 0

```
R1>enable
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto ipsec transform-set VPN-SET esp-esp esp-sha-hmac
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
R1(config)#
```

Copy Paste

ROUTER 1

```
R2>enable
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#crypto ipsec transform-set VPN-SET esp-esp esp-sha-hmac
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#
```

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Configure Crypto Map on R1

```
R1>enable
R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto ipsec transform-set VPN-SET esp-esp esp-sha-hmac
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
R1(config)#config terminal
%Invalid hex value
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R1(config-crypto-map)#set peer 10.0.0.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#
```

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Configure Crypto Map on R2

```
R2>enable
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#crypto ipsec transform-set VPN-SET esp-esp esp-sha-hmac
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R2(config-crypto-map)#set peer 10.0.0.1
R2(config-crypto-map)#set transform-set VPN-SET
R2(config-crypto-map)#match address 110
R2(config-crypto-map)#exit
R2(config)#
```

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Apply Crypto Map to R1

```
R1(config)#interface serial 0/3/0
R1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#
```

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Apply Crypto Map to R2

```
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R2(config-crypto-map)#set peer 10.0.0.1
R2(config-crypto-map)#set transform-set VPN-SET
R2(config-crypto-map)#match address 110
R2(config-crypto-map)#exit
R2(config)#interface serial 0/3/0
R2(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R2(config-if)#exit
R2(config)#end
R2#
MVS-S-CONFIG I: Configured from console by console
write memory
Building configuration...
[OK]
R2#
```

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Router 1 crypto

```
Building configuration...
[OK]
R1#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
  Peer = 10.0.0.2
  Extended IP access list 110
    access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
  Current peer: 10.0.0.2
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (V3): N
  Transform set=
    VPN-SET,
  Interfaces using crypto map VPN-MAP:
    Serial0/3/0
R1#
```

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Router 2 crypto

```
Building configuration...
[OK]
R2#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
  Peer = 10.0.0.1
  Extended IP access list 110
    access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
  Current peer: 10.0.0.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (V3): N
  Transform set=
    VPN-SET,
  Interfaces using crypto map VPN-MAP:
    Serial0/3/0
R2#
```

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Generate IPsec Traffic

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 1ms
C:\>
```

Check IKE/ISAKMP on R1

Check IPsec on R1

```
R2#show crypto ipsec sa
interface: Serial0/3/0
  crypto map tag: VPN-MAP, local addr 10.0.0.2
    protected vrf: (none)
    local ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
    remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
    current peer 10.0.0.1 port 500
    PERMIT, flags=(origin_is_acl,):
    #pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
    #pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompr. failed: 0
    #send errors 0, #recv errors 0
    local crypto endpt.: 10.0.0.2, remote crypto endpt.: 10.0.0.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/3/0
    current outbound spi: 0x557F5F8A(48505489)
    inbound esp sa:
      spi: 0x6553F65D(1753476749)
      transform: esp-esp esp-sha-hmac
R2#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dat      src      state      conn-id slot status
10.0.0.1  10.0.0.2    QM_IDLE   1027    0 ACTIVE
IPv6 Crypto ISAKMP SA
R2#
```

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Verify IPsec on R2

```
Serial0/3/0
R2#show crypto ipsec sa
interface: Serial0/3/0
Crypto map tag: VPN-MAP, local addr 10.0.0.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
current_peer 10.0.0.1 port 500
PERMIT, flags={origin_is_acl,}
#pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
#pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 10.0.0.2, remote crypto endpt.: 10.0.0.1
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/3/0
current outbound spi: 0x1557EF64(358084464)

inbound esp sa:
spi: 0x6833F63D(1753478749)
transform: esp-aes esp-sha-hmac ,

R2#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          conn-id  slot  status
10.0.0.1      10.0.0.2      QM_IDLE    1027    0 ACTIVE

IPv6 Crypto ISAKMP SA

R2#
```

Verify the Crypto Map R0

```
R1#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.2
Extended IP access list 110
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Current peer: 10.0.0.2
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Y/N): N
Transform set=
VPN-SET,
Interfaces using crypto map VPN-MAP:
Serial0/3/0

R1#
```

R1

```
R2#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.1
Extended IP access list 110
access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Current peer: 10.0.0.1
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Y/N): N
Transform set=
VPN-SET,
Interfaces using crypto map VPN-MAP:
Serial0/3/0

R2#
```

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Final Test

From PC1:

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Reply from 192.168.1.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 0ms
C:\>
```

From PC0:

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```

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